bing you of sleep—screen time. Social media and OTT were the reasons 54 per cent of the study's respondents stayed up late; 88 per cent had used their phones just before going to bed. This results in a double whammy of delayed and disturbed sleep. "Blue light fools your brain into thinking it is daytime and your body stops releasing melatonin," says Dr Sundaram. "This reduction stops our natural wind-down process and delays sleep." And even after we go to sleep, those who have spent too much time on phones or accessing media tend to sleep light. "Phones keep your brain awake by continuously stimulating it. It is the most common cause of sleep disturbance even after you have fallen asleep," says Dr Ashish Kumar Prakash, consultant, respiratory and sleep medicine, Medanta, Gurugram. The Wakefit study corroborates this—88 per cent of its participants reported waking up multiple times during the night.

Stress, too, is a factor in determining our sleep quantity and quality. "We don't get enough time to relax and wind down at night," says Dr Suri. "Our brains are highly aroused from work or entertainment or worries about the future, leading to difficulty falling asleep. Stress is particularly associated with sleep initiation (sleep onset insomnia). Some people resort to alcohol for falling asleep. However, while it may help them go to sleep, it impairs the depth and quality of sleep."

Finally, the combination of unhealthy diets and sedentary lifestyles has led to a rise in obesity rates, which in turn has become a leading cause of the sleep disorder known as obstructive sleep apnea. A 2024 global analysis, published in the medical journal *The Lancet*, found 44 million women and 26 million men aged above 20 in India to be obese, up from 2.4 million women and 1.1 million men in 1990. The fat deposits around the upper airway tend to block your breathing and cause sleep apnea. And it's not just adults who are afflicted with it. According to Pune-based sleep specialist Dr Amrita Jain, "Today, because of obesity, I get even children as patients. That said, the number of people who come for treatment still remains very low." Sleep apnea can lead to daytime sleepiness and reduce cognitive performance over time. Other causes of poor sleep include sleepwalking, nightmares, sleep-related eating disorders, teeth grinding, jet lag and working late night shifts.



"Phones keep your brain awake by continuously stimulating it. It is the most common cause of sleep disturbance even after you have fallen asleep"

DR A.K. PRAKASH
Consultant, Respiratory & Sleep Medicine, Medanta, Gurugram

What does sleep debt do to your health?

Sleep is essential for nearly every bodily function. "Poor sleep or sleeping for less than six hours can predispose one to conditions like hypertension, diabetes, coronary heart disease, stroke, depression, and even premature death. Furthermore, inadequate sleep impairs our cognitive functioning, economic productivity and quality of life," says Dr Suri.

A study conducted by the National Institutes of Health, US, in collaboration with Google and Fitbit's sleep-debt data, and published in the journal *Nature Medicine* in July 2024, analysed the sleep patterns of 6,785 participants in the US. It found that insufficient sleep quantity, quality and regularity were all associated with increased incidence of numerous chronic problems, including obesity, irregular heartbeat, hypertension, major depressive disorder, increased blood cholesterol levels and generalised anxiety disorder.

New research is also shedding light on some unusual impacts. A 2021 study in the journal *Scientific Reports* found that sleep deprivation impacted gait and the sense of balance. Another 2024 study in the journal *Neurology* concluded that disrupted sleep in your 30s and 40s could double your odds for cognitive decline in the future. "Just like poor mental health can affect sleep, poor sleep can lead to quite significant mental health conditions like anxiety, depression, cognitive impairment, or problems with attention, concentration and memory, including incidences of psychosis," says Dr Ramaiah. Ongoing research is also investigating how sleep debt weakens the body's immune system, leads to weight gain and issues with male and female fertility. "Irregular sleeping patterns adversely impact sperm quality and quantity in men. In women, a proper circadian rhythm aids the secretion of certain good hormones; if that is off, then hormones like LH, ovulation, progesterone go haywire," says Dr Sandeep Talwar, fertility specialist at NOVA IVF.

Sleep debt also increases the likelihood of accidents. "Eventually, no matter how much you push yourself to stay awake, tiredness catches up with you. It is not uncommon for drivers to engage in microsleep episodes where you doze off for a few seconds, and be in a fatal accident," says Dr Suri. Each year, drowsy driving accounts for about 100,000 crashes, 71,000 injuries and 1,550 fatalities, according to a National Safety Council report referenced in a 2021 paper of The International Journal of Analytical and Experimental Modal Analysis (IJAEMA).